

Chatbot_Arena_Leaderboard_Calculation_(Bradley_Terry_model)

April 28, 2026

1 Introduction

In this notebook, we present data analysis on Chatbot Arena data collected from <https://arena.lmsys.org>.

We explain different Elo calculation methods (online Elo and Bradley-Terry model) for model ranking.

To view the latest leaderboard, see <https://huggingface.co/spaces/lmsys/chatbot-arena-leaderboard>.

```
[1]: import sys
    !{sys.executable} -m pip install gdown numpy pandas plotly tqdm requests
    ↪ ipython nbformat

from collections import defaultdict
import json, math, gdown
import numpy as np
import pandas as pd
import plotly.express as px
from tqdm import tqdm
import requests
pd.options.display.float_format = '{:.2f}'.format
```

```
Requirement already satisfied: gdown in
/home/madison/.mypipenv/lib/python3.14/site-packages (6.0.0)
Requirement already satisfied: numpy in
/home/madison/.mypipenv/lib/python3.14/site-packages (2.4.3)
Requirement already satisfied: pandas in
/home/madison/.mypipenv/lib/python3.14/site-packages (3.0.1)
Requirement already satisfied: plotly in
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Requirement already satisfied: requests in
/home/madison/.mypipenv/lib/python3.14/site-packages (2.32.5)
Requirement already satisfied: ipython in
```

```

/home/madison/.mypipenv/lib/python3.14/site-packages (9.11.0)
Collecting nbformat
  Using cached nbformat-5.10.4-py3-none-any.whl.metadata (3.6 kB)
Requirement already satisfied: beautifulsoup4 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from gdown) (4.14.3)
Requirement already satisfied: filelock in
/home/madison/.mypipenv/lib/python3.14/site-packages (from gdown) (3.29.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: narwhals>=1.15.1 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from plotly) (2.18.0)
Requirement already satisfied: packaging in
/home/madison/.mypipenv/lib/python3.14/site-packages (from plotly) (26.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from requests) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in
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Requirement already satisfied: urllib3<3,>=1.21.1 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from requests) (2.6.3)
Requirement already satisfied: certifi>=2017.4.17 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from requests) (2026.2.25)
Requirement already satisfied: decorator>=5.1.0 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from ipython) (5.2.1)
Requirement already satisfied: ipython-pygments-lexers>=1.0.0 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from ipython) (1.1.1)
Requirement already satisfied: jedi>=0.18.2 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from ipython) (0.19.2)
Requirement already satisfied: matplotlib-inline>=0.1.6 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from ipython) (0.2.1)
Requirement already satisfied: pexpect>4.6 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from ipython) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from ipython) (3.0.52)
Requirement already satisfied: pygments>=2.14.0 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from ipython) (2.19.2)
Requirement already satisfied: stack_data>=0.6.0 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from ipython) (0.6.3)
Requirement already satisfied: traitlets>=5.13.0 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from ipython) (5.14.3)
Requirement already satisfied: wcwidth in
/home/madison/.mypipenv/lib/python3.14/site-packages (from
prompt_toolkit<3.1.0,>=3.0.41->ipython) (0.6.0)
Collecting fastjsonschema>=2.15 (from nbformat)
  Using cached fastjsonschema-2.21.2-py3-none-any.whl.metadata (2.3 kB)
Collecting jsonschema>=2.6 (from nbformat)
  Downloading jsonschema-4.26.0-py3-none-any.whl.metadata (7.6 kB)
Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from nbformat) (5.9.1)

```

Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from
jedi>=0.18.2->ipython) (0.8.6)

Collecting attrs>=22.2.0 (from jsonschema>=2.6->nbformat)
Using cached attrs-26.1.0-py3-none-any.whl.metadata (8.8 kB)

Collecting jsonschema-specifications>=2023.03.6 (from jsonschema>=2.6->nbformat)
Using cached jsonschema_specifications-2025.9.1-py3-none-any.whl.metadata (2.9
kB)

Collecting referencing>=0.28.4 (from jsonschema>=2.6->nbformat)
Using cached referencing-0.37.0-py3-none-any.whl.metadata (2.8 kB)

Collecting rpds-py>=0.25.0 (from jsonschema>=2.6->nbformat)
Using cached rpds_py-0.30.0-cp314-cp314-
manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (4.1 kB)

Requirement already satisfied: platformdirs>=2.5 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from jupyter-
core!=5.0.*,>=4.12->nbformat) (4.9.4)

Requirement already satisfied: ptyprocess>=0.5 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from pexpect>4.6->ipython)
(0.7.0)

Requirement already satisfied: six>=1.5 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from python-
dateutil>=2.8.2->pandas) (1.17.0)

Requirement already satisfied: executing>=1.2.0 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from
stack_data>=0.6.0->ipython) (2.2.1)

Requirement already satisfied: asttokens>=2.1.0 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from
stack_data>=0.6.0->ipython) (3.0.1)

Requirement already satisfied: pure-eval in
/home/madison/.mypipenv/lib/python3.14/site-packages (from
stack_data>=0.6.0->ipython) (0.2.3)

Requirement already satisfied: soupsieve>=1.6.1 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from
beautifulsoup4->gdown) (2.8.3)

Requirement already satisfied: typing-extensions>=4.0.0 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from
beautifulsoup4->gdown) (4.15.0)

Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in
/home/madison/.mypipenv/lib/python3.14/site-packages (from
requests[socks]->gdown) (1.7.1)

Using cached nbformat-5.10.4-py3-none-any.whl (78 kB)

Using cached fastjsonschema-2.21.2-py3-none-any.whl (24 kB)

Downloading jsonschema-4.26.0-py3-none-any.whl (90 kB)

Using cached attrs-26.1.0-py3-none-any.whl (67 kB)

Using cached jsonschema_specifications-2025.9.1-py3-none-any.whl (18 kB)

Using cached referencing-0.37.0-py3-none-any.whl (26 kB)

Using cached
rpds_py-0.30.0-cp314-cp314-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (382

kB)

Installing collected packages: fastjsonschema, rpds-py, attrs, referencing, jsonschema-specifications, jsonschema, nbformat

7/7 [nbformat] 5/7 [jsonschema]

Successfully installed attrs-26.1.0 fastjsonschema-2.21.2

jsonschema-4.26.0 jsonschema-specifications-2025.9.1 nbformat-5.10.4

referencing-0.37.0 rpds-py-0.30.0

[notice] A new release of pip is

available: 25.3 -> 26.0.1

[notice] To update, run:

/home/madison/.mypipenv/bin/python -m pip install --upgrade pip

2 Obtaining and Cleaning the Tournament Data

We are hosting the initial tournament results as a JSON file on Google Drive. We use the `gdown` function to download the data. The data contains all the battles and voting results collected for ranking models.

```
[2]: # we use the latest data
url = "https://storage.googleapis.com/arena_external_data/public/
      ↪clean_battle_20240814_public.json"
response = requests.get(url)

with open('local_file_name.json', 'wb') as file:
    file.write(response.content)

# load the JSON data from the local file
with open('local_file_name.json', 'r') as file:
    battles = pd.read_json(file).sort_values(ascending=True, by=["tstamp"])
```

```
[3]: battles
```

```
[3]:
```

	model_a	model_b	winner \
0	chatglm-6b	koala-13b	model_b
1	oasst-pythia-12b	alpaca-13b	tie
2	koala-13b	oasst-pythia-12b	model_b
3	vicuna-13b	oasst-pythia-12b	model_b
4	vicuna-13b	koala-13b	model_a
...	...	...	...
1799986	llama-3-8b-instruct	chatgpt-4o-latest	model_a
1799987	gpt-4o-2024-08-06	claude-3-5-sonnet-20240620	model_a
1799988	llama-3.1-8b-instruct	reka-flash-20240722	tie (bothbad)
1799989	gemini-1.5-pro-api-0514	claude-3-haiku-20240307	model_a
1799990	llama-3.1-405b-instruct	claude-3-5-sonnet-20240620	tie (bothbad)

	judge	turn	anony	language	tstamp \
0	2e9c29aa140b8e50643235eab01dc9ea	1	True	English	1682351591.13
1	2e9c29aa140b8e50643235eab01dc9ea	1	True	English	1682351654.67
2	2e9c29aa140b8e50643235eab01dc9ea	1	True	English	1682351708.94
3	2e9c29aa140b8e50643235eab01dc9ea	1	True	English	1682351785.19
4	2e9c29aa140b8e50643235eab01dc9ea	1	True	English	1682351891.66
...	...	...	...	...	...
1799986	1f08bb8a2f8f77ef1dd3e2f91c4214f2	1	True	English	1723665137.57
1799987	211cc2f9ea2b5846cfa3c373f32db8f4	1	True	Russian	1723665139.77
1799988	1bf5cd833334692589920215e6a5ad84	1	True	Polish	1723665143.79
1799989	41893b64a23df36ec2e2acbc7aa78def	2	True	English	1723665163.09
1799990	1f08bb8a2f8f77ef1dd3e2f91c4214f2	1	True	English	1723665173.41

	conv_metadata	is_code \
0	{'sum_user_tokens': 10, 'sum_assistant_a_token...	True
1	{'sum_user_tokens': 11, 'sum_assistant_a_token...	False
2	{'sum_user_tokens': 10, 'sum_assistant_a_token...	False
3	{'sum_user_tokens': 9, 'sum_assistant_a_tokens...	False
4	{'sum_user_tokens': 5, 'sum_assistant_a_tokens...	False
...	...	...
1799986	{'sum_user_tokens': 18, 'sum_assistant_a_token...	False
1799987	{'sum_user_tokens': 436, 'sum_assistant_a_toke...	False
1799988	{'sum_user_tokens': 28, 'sum_assistant_a_token...	False
1799989	{'sum_user_tokens': 42, 'sum_assistant_a_token...	False
1799990	{'sum_user_tokens': 8, 'sum_assistant_a_tokens...	False

	is_refusal	dedup_tag \
0	False	{'high_freq': False, 'sampled': True}
1	False	{'high_freq': False, 'sampled': True}
2	False	{'high_freq': False, 'sampled': True}
3	False	{'high_freq': False, 'sampled': True}
4	True	{'high_freq': False, 'sampled': True}
...	...	...
1799986	False	{'high_freq': False, 'sampled': True}
1799987	False	{'high_freq': False, 'sampled': True}
1799988	False	{'high_freq': False, 'sampled': True}
1799989	False	{'high_freq': False, 'sampled': True}
1799990	True	{'high_freq': False, 'sampled': True}

	category_tag
0	{'if_v0.1': {'if': False, 'score': 1}, 'math_v...
1	{'if_v0.1': {'if': False, 'score': 0}, 'math_v...
2	{'if_v0.1': {'if': False, 'score': 0}, 'math_v...
3	{'if_v0.1': {'if': False, 'score': 1}, 'math_v...
4	{'if_v0.1': {'if': False, 'score': 0}, 'math_v...
...	...
1799986	{'if_v0.1': {'if': False, 'score': 1}, 'math_v...

```

1799987 {'if_v0.1': {'if': False, 'score': 3}, 'math_v...
1799988 {'if_v0.1': {'if': False, 'score': 2}, 'math_v...
1799989 {'if_v0.1': {'if': False, 'score': 1}, 'math_v...
1799990 {'if_v0.1': {'if': False, 'score': 1}, 'math_v...

```

```
[1799991 rows x 13 columns]
```

```

[4]: # we use anony battles only for leaderboard
battles = battles[battles["anony"] == True]

# we de-duplicate top 0.1% redudant prompts
# see https://lmsys.org/blog/2024-05-17-category-hard/
↳#note-enhancing-quality-through-de-duplication
print("Before dedup: ", len(battles))
battles = battles[battles["dedup_tag"].apply(lambda x: x.get("sampled", False))]
print("After dedup: ", len(battles))

```

Before dedup: 1799991

After dedup: 1670250

3 Exploratory Analysis

Before computing the Elo ratings, we first conduct some basic exploratory analysis to highlight a few key properties and caveates with this data.

3.1 Statistics

We allowed the user to declare a tie between the pairs of models. To collect additional data, later in the tournament we also allowed the user to declare a tie in which both models were bad. There were a significant portion of tied outcomes.

```

[5]: fig = px.bar(battles["winner"].value_counts(),
                  title="Counts of Battle Outcomes", text_auto=True, height=400)
fig.update_layout(xaxis_title="Battle Outcome", yaxis_title="Count",
                  showlegend=False)
fig

```

A bar chart titled 'Battle Outcome' showing the count of outcomes for four categories: model_b, model_a, tie (bothbad), and tie. The y-axis is labeled 'Count' and ranges from 0 to 500k. The x-axis is labeled 'Battle Outcome'. The bars are blue. The counts are: model_b (550.81k), model_a (543.065k), tie (bothbad) (295.254k), and tie (281.121k).

Battle Outcome	Count
model_b	550.81k
model_a	543.065k
tie (bothbad)	295.254k
tie	281.121k

```
[6]: battles_no_ties = battles[~battles["winner"].str.contains("tie")]
```

3.2 Non-uniform Model Frequency

The model frequency is not uniform because of the following reasons: - Several different matching and sampling algorithms were used. We employed uniform sampling as well as weighted sampling methods, which assign greater weights to better models. - Some new models were added later.

```
[7]: fig = px.bar(pd.concat([battles["model_a"], battles["model_b"]]).value_counts(),
                  title="Battle Count for Each Model", text_auto=True)
fig.update_layout(xaxis_title="model", yaxis_title="Battle Count", height=400,
                  showlegend=False)
fig
```

model	Battle Count (approx.)
dolphin-2.2.1-mistral-7b	145,000
zephyr-7b-alpha	140,000
mpt-30b-chat	135,000
gpt4o-35b	130,000
nous-hermes-2	125,000
do-1.5-72b	120,000
zephyr-7b-beta	115,000
zephyr-orpo-1.41b-A35b-v0.1	110,000
llm-ai-chat-5.3b	105,000
oasst-pythia-12b	100,000
gemini-pro	95,000
llm-ai-3b-chat	90,000
llm-ai-7b	85,000
reka-flash-20240722	80,000
codellama-34b-instruct	75,000
qwen1.5-72b-chat	70,000
llm-ai-70b-instruct	65,000
gpt-4o-2024-08-06	60,000
sterling-im-7b-alpha	55,000
gemma-2-2b-it	50,000
gemma-1.1-2b-it	45,000
mistral-large-2407	40,000
mistral-next	35,000
claude-2.0	30,000
llama-3.1-70b-instruct	25,000
llama-2-7b-chat	20,000
deepseek-v2-api-0628	15,000
y-34b-chat	10,000
pi-3-small-8k-instruct	5,000
gemini-pro-v0.5p	4,000
gpt-4o-2024-07-18	3,000
gemma-13b	2,000
mistral-7b-instruct-v0.2	1,500
claude-instant-1	1,000
pi-3-mini-4k-instruct	500
qwen1.5-32b-chat	400
gemma-1.1-7b-it	300
gemma-2-9b-it	200
reka-flash-21b-20240226	150
gemma-2-27b-it	100
snowflake-arctic-instruct	50
mistral-medium	40
gpt-3.5-turbo-0613	30
qwen1.5-72b-chat	20
mistral-8x22b-instruct-v0.1	15
gpt-4o-mini-0514	10
command-r	5
mistral-large-2402	4
gemini-1.5-pro-gp-0514	3
gpt-4o-2024-05-13	2
gpt-4-turbo-2024-04-09	1
gpt-4-0613	1
llama-3.8b-instruct	1
claude-3-sonnet-20240229	1
llama-3-70b-instruct	1

We examine the number of pairings for each combination of models.

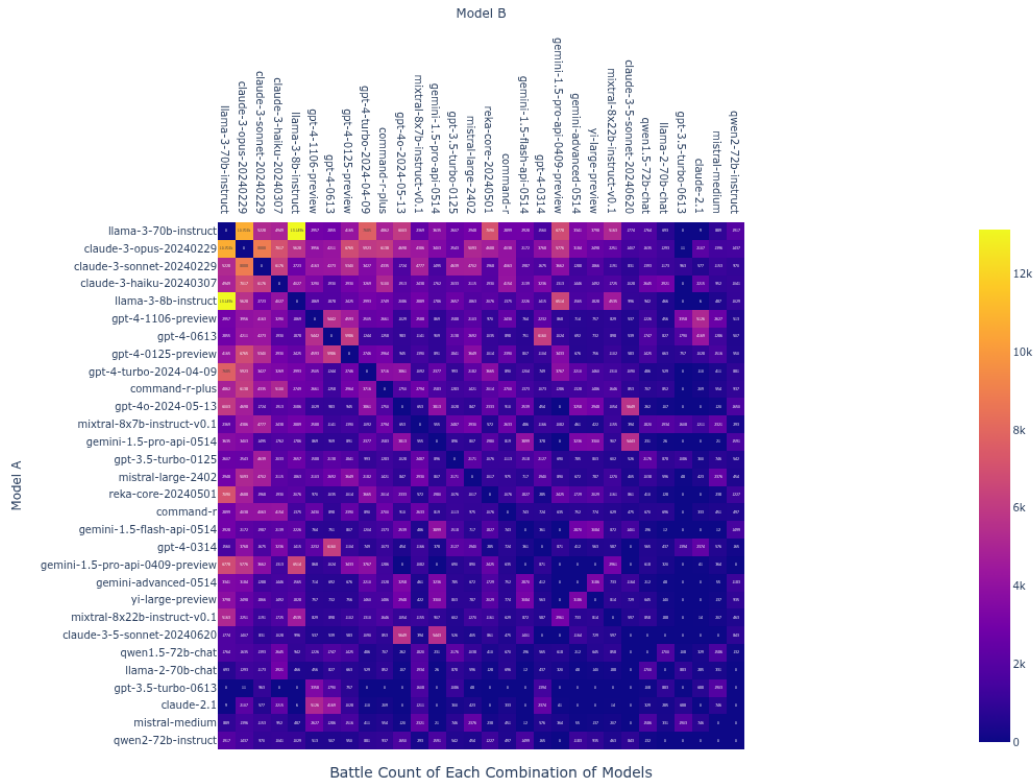
```
[8]: def visualize_battle_count(battles, title, show_num_models=30):
```

```

ptbl = pd.pivot_table(battles, index="model_a", columns="model_b",
    aggfunc="size",
    fill_value=0)
battle_counts = ptbl + ptbl.T
ordering = battle_counts.sum().sort_values(ascending=False).index
ordering = ordering[:show_num_models]
fig = px.imshow(battle_counts.loc[ordering, ordering],
    title=title, text_auto=True)
fig.update_layout(xaxis_title="Model B",
    yaxis_title="Model A",
    xaxis_side="top", height=800, width=800,
    title_y=0.07, title_x=0.5,
    font=dict(size=10))
fig.update_traces(hovtemplate=
    "Model A: %{y}<br>Model B: %{x}<br>Count: %{z}<extra></
    extra>")
return fig

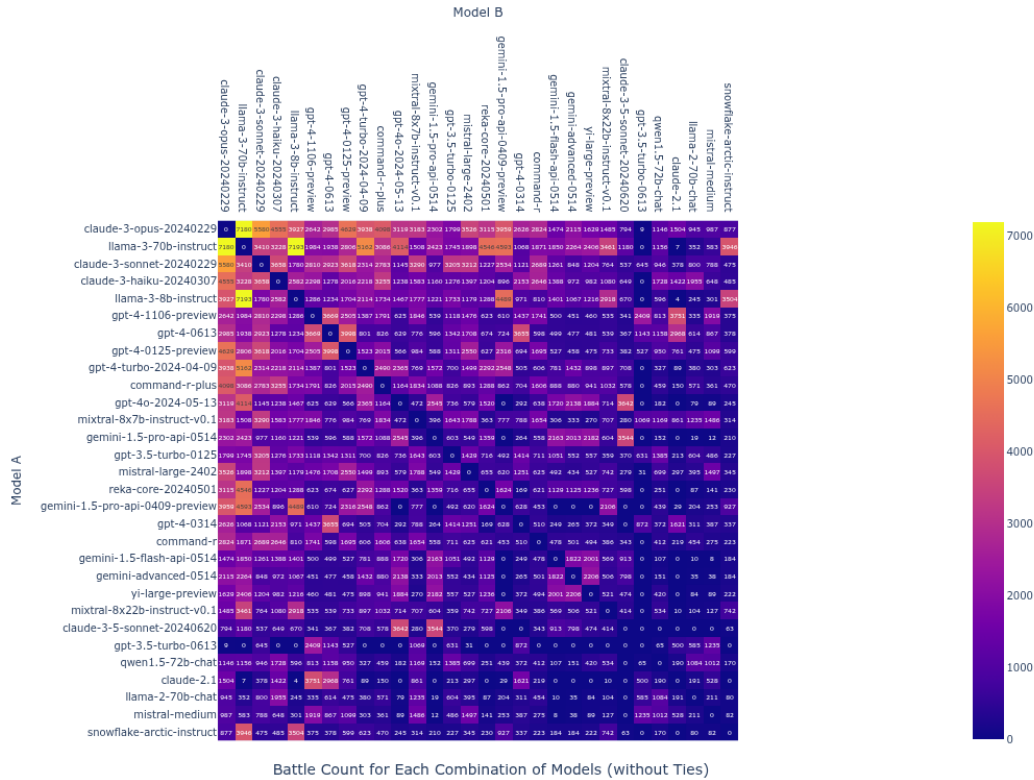
fig = visualize_battle_count(battles, title="Battle Count of Each Combination
    of Models", show_num_models=30)
fig

```



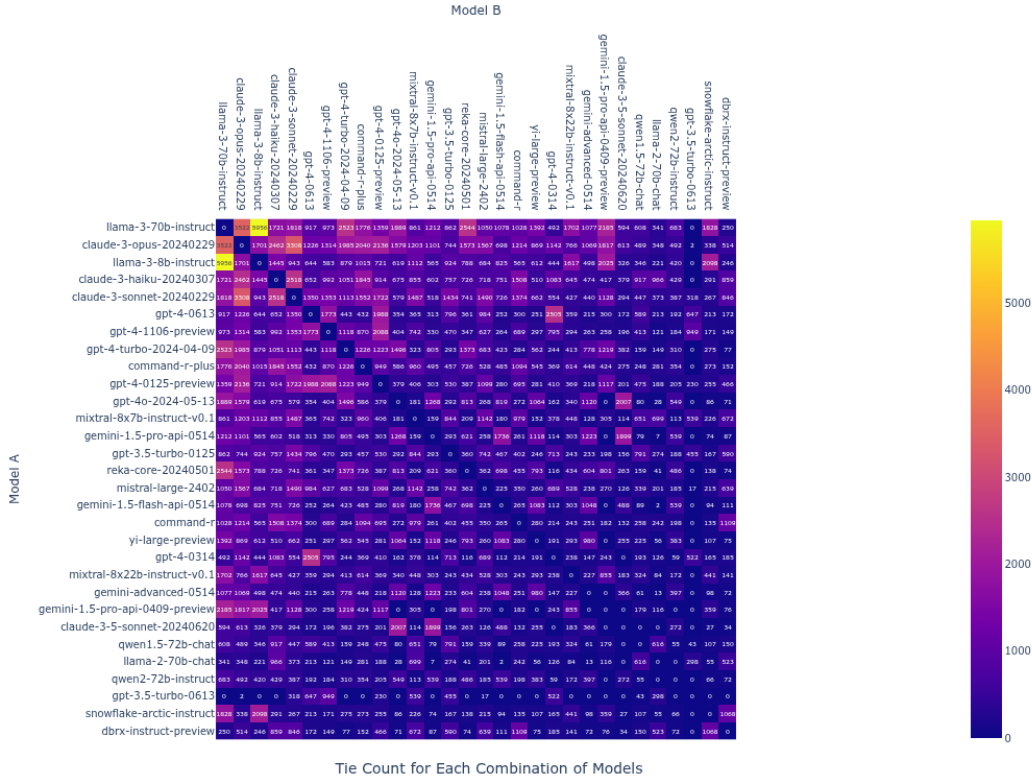
3.2.1 Battles Excluding Ties

[9]: visualize_battle_count(battles_no_ties, "Battle Count for Each Combination of_↵Models (without Ties)")



3.2.2 Counting Ties

[10]: visualize_battle_count(battles[battles['winner'].str.contains("tie")], "Tie_↵Count for Each Combination of Models")



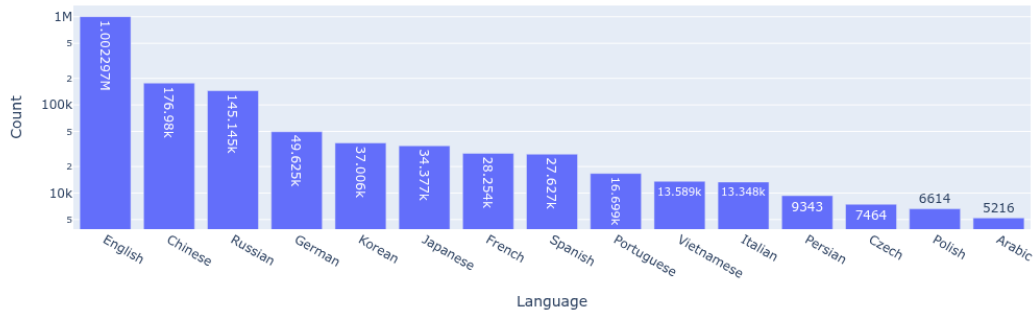
3.3 Inferred Language

We also inferred the language for each conversation using polyglot package. This is just an estimate but will help guide future analysis. The vast majority of conversations were in English.

```
[11]: lang_count = battles["language"].value_counts()
lang_count = lang_count.drop(index=("unknown"))
```

```
[12]: topk = 15
fig = px.bar(lang_count.head(topk),
             title=f"Battle Counts for the Top {topk} Languages",
             text_auto=True, height=400, log_y=True)
fig.update_layout(xaxis_title="Language", yaxis_title="Count", showlegend=False)
fig
```

Battle Counts for the Top 15 Languages

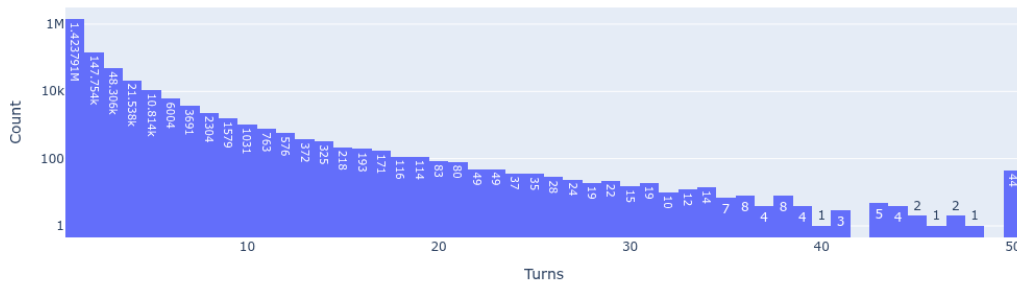


3.4 Number of Conversation Turns

We also noticed that most conversations only have one turn.

```
[13]: fig = px.histogram(battles["turn"],
                        title=f"Number of Conversation Turns",
                        text_auto=True, height=400, log_y=True)
fig.update_layout(xaxis_title="Turns", yaxis_title="Count", showlegend=False)
fig
```

Number of Conversation Turns



3.5 Pairwise Win Fractions

Finally, we can also compute the pairwise win fractions. However, because each model can play as Model A and as Model B and win in both situations we need to compute the wins in both configurations divided by the number of pairings of each model.

```
[14]: def compute_pairwise_win_fraction(battles, max_num_models=30):
        # Times each model wins as Model A
```

```

a_win_ptbl = pd.pivot_table(
    battles[battles['winner'] == "model_a"],
    index="model_a", columns="model_b", aggfunc="size", fill_value=0)

# Table counting times each model wins as Model B
b_win_ptbl = pd.pivot_table(
    battles[battles['winner'] == "model_b"],
    index="model_a", columns="model_b", aggfunc="size", fill_value=0)

# Table counting number of A-B pairs
num_battles_ptbl = pd.pivot_table(battles,
    index="model_a", columns="model_b", aggfunc="size", fill_value=0)

# Computing the proportion of wins for each model as A and as B
# against all other models
row_beats_col_freq = (
    (a_win_ptbl + b_win_ptbl.T) /
    (num_battles_ptbl + num_battles_ptbl.T)
)

# Arrange ordering according to proportion of wins
prop_wins = row_beats_col_freq.mean(axis=1).sort_values(ascending=False)
prop_wins = prop_wins[:max_num_models]
model_names = list(prop_wins.keys())
row_beats_col = row_beats_col_freq.loc[model_names, model_names]
return row_beats_col

def visualize_pairwise_win_fraction(battles, title, max_num_models=30):
    row_beats_col = compute_pairwise_win_fraction(battles, max_num_models)
    fig = px.imshow(row_beats_col, color_continuous_scale='RdBu',
        text_auto=".2f", title=title)
    fig.update_layout(xaxis_title="Model B: Loser",
        yaxis_title="Model A: Winner",
        xaxis_side="top", height=900, width=900,
        title_y=0.07, title_x=0.5)
    fig.update_traces(hovertemplate=
        "Model A: %{y}<br>Model B: %{x}<br>Fraction of A Wins:␣
        ↳%{z}<extra></extra>")

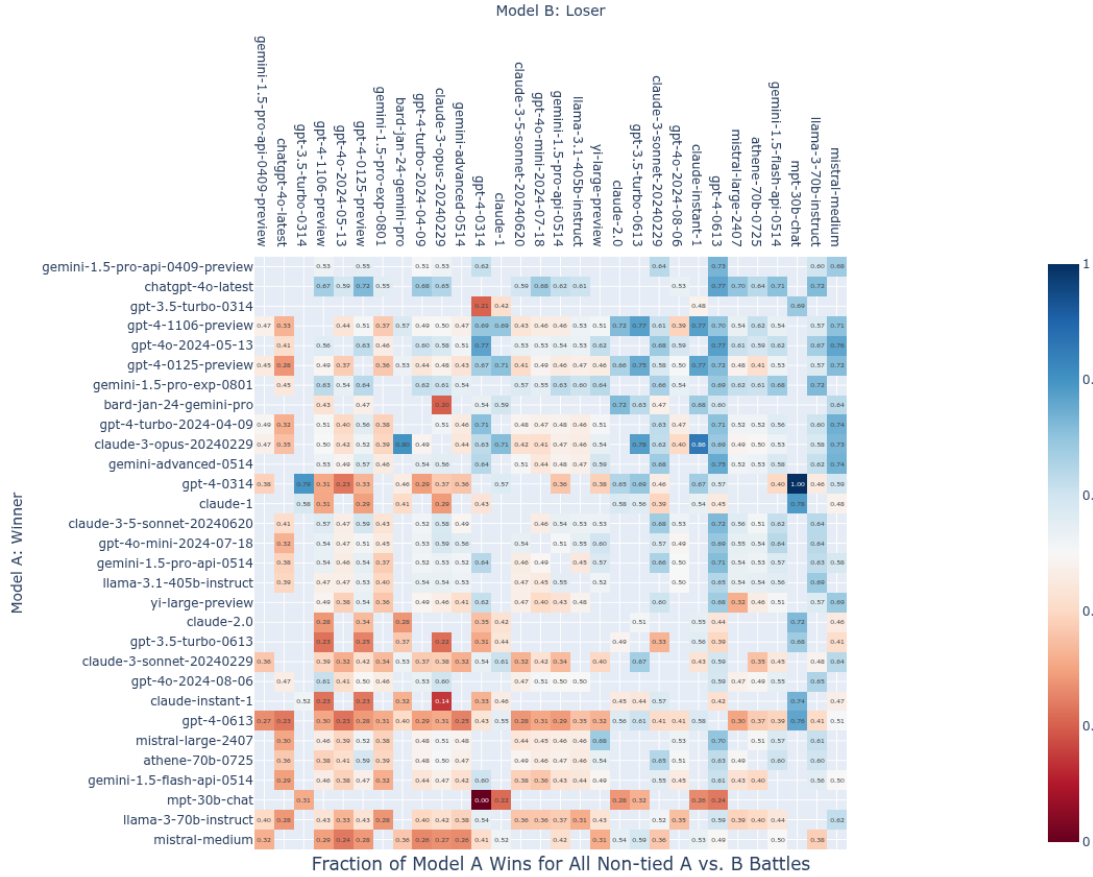
    return fig

```

```

[15]: fig = visualize_pairwise_win_fraction(battles_no_ties,
    title = "Fraction of Model A Wins for All Non-tied A vs. B Battles")
fig

```

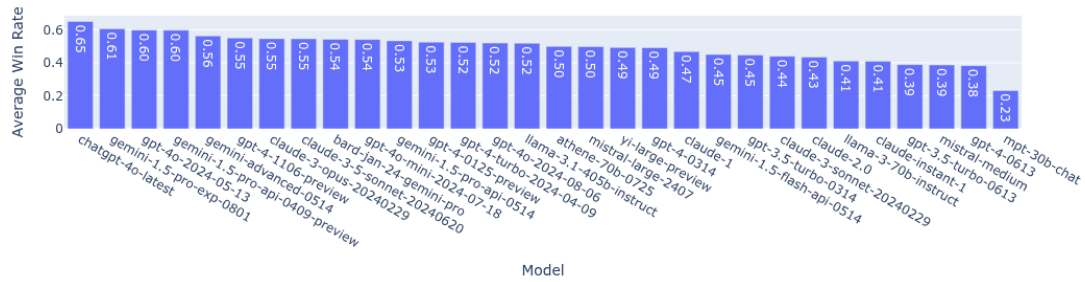


3.6 Preliminary Ranking

Using just the average win rate against all other models we can already compute an estimated leaderboard. However, this method may not be as scalable as the Elo rating system that we will use later because this method requires data from all model combinations.

```
[16]: row_beats_col_freq = compute_pairwise_win_fraction(battles_no_ties)
fig = px.bar(row_beats_col_freq.mean(axis=1).sort_values(ascending=False),
             title="Average Win Rate Against All Other Models (Assuming Uniform
             ↳Sampling and No Ties)",
             text_auto=".2f")
fig.update_layout(yaxis_title="Average Win Rate", xaxis_title="Model",
                  showlegend=False)
fig
```

Average Win Rate Against All Other Models (Assuming Uniform Sampling and No Ties)



#Elo Ratings

The [Elo rating system](#) is a method for calculating the relative skill levels of players, which has been widely adopted in chess and other competitive games. The difference in the ratings between two players serves as a predictor of the outcome of a match. The Elo rating system works well for our case because we have multiple models and we run pairwise battles between them. In this section, we present different methods for calculating Elo ratings.

3.6.1 Compute Ratings

We first use the online linear update algorithm to compute Elo ratings. We choose a small K-factor of 4 to make the Elo ratings more stable and less biased towards recent games.

```
[17]: def compute_online_elo(battles, K=4, SCALE=400, BASE=10, INIT_RATING=1000):
    rating = defaultdict(lambda: INIT_RATING)

    for rd, model_a, model_b, winner in battles[['model_a', 'model_b', 'winner']].itertuples():
        ra = rating[model_a]
        rb = rating[model_b]
        ea = 1 / (1 + BASE ** ((rb - ra) / SCALE))
        eb = 1 / (1 + BASE ** ((ra - rb) / SCALE))
        if winner == "model_a":
            sa = 1
        elif winner == "model_b":
            sa = 0
        elif winner == "tie" or winner == "tie (bothbad)":
            sa = 0.5
        else:
            raise Exception(f"unexpected vote {winner}")
        rating[model_a] += K * (sa - ea)
        rating[model_b] += K * (1 - sa - eb)

    # calibrate llama-13b to 800
    delta = (800 - rating["llama-13b"])
```

```

for model in battles["model_a"].unique():
    rating[model] += delta

return rating

```

```

[18]: def pretty_print_model_ratings(ratings):
    df = pd.DataFrame([
        [n, ratings[n]] for n in ratings.keys()
    ], columns=["Model", "Elo rating"]).sort_values("Elo rating",
↪ascending=False).reset_index(drop=True)
    # df["Elo rating"] = (df["Elo rating"] + 0.5).astype(int)
    df.index = df.index + 1
    return df

online_elo_ratings = compute_online_elo(battles)
pretty_print_model_ratings(online_elo_ratings)

```

```

[18]:

```

	Model	Elo rating
1	gemini-1.5-pro-api-0409-preview	1106.87
2	gemini-1.5-pro-exp-0801	1074.94
3	chatgpt-4o-latest	1073.74
4	gpt-3.5-turbo-0314	1051.02
5	bard-jan-24-gemini-pro	1041.26
..	...	...
125	chatglm3-6b	814.55
126	oasst-pythia-12b	812.39
127	llama-13b	800.00
128	fastchat-t5-3b	794.37
129	dolly-v2-12b	781.44

[129 rows x 2 columns]

However, even with a small K-factor, we still found this online update algorithm to be unstable.

To demonstrate it, we recompute Elo rating by using the reversed game order and observe significant difference due to online update of Elo which biases the recent games.

```

[19]: def pretty_print_two_ratings(ratings_1, ratings_2, column_names):
    df = pd.DataFrame([
        [n, ratings_1[n], ratings_2[n]] for n in ratings_1.keys()
    ], columns=["Model", column_names[0], column_names[1]]).
↪sort_values(column_names[0], ascending=False).reset_index(drop=True)
    df[column_names[0]] = (df[column_names[0]] + 0.5).astype(int)
    df[column_names[1]] = (df[column_names[1]] + 0.5).astype(int)
    df.index = df.index + 1
    return df

elo_mle_ratings_reverse = compute_online_elo(battles.iloc[::-1])

```

```
preety_print_two_ratings(online_elo_ratings,
                        elo_mle_ratings_reverse,
                        column_names=["Elo rating", "Elo rating with reverse_
↳order"])
```

```
[19]:
```

	Model	Elo rating \
1	gemini-1.5-pro-api-0409-preview	1107
2	gemini-1.5-pro-exp-0801	1075
3	chatgpt-4o-latest	1074
4	gpt-3.5-turbo-0314	1051
5	bard-jan-24-gemini-pro	1041
..	...	...
125	chatglm3-6b	815
126	oasst-pythia-12b	812
127	llama-13b	800
128	fastchat-t5-3b	794
129	dolly-v2-12b	781

	Elo rating with reverse order
1	1112
2	1086
3	1151
4	1162
5	1155
..	...
125	893
126	907
127	800
128	863
129	833


```
[129 rows x 3 columns]
```

3.6.2 Bradley-Terry model

In the context of LLM evaluation, models can be assumed to be static. In this case, we can directly fit the ratings with Bradley-Terry model, which produce significantly stable ratings. Here we provide an implementation with logistic regression.

```
[20]: def compute_mle_elo(
df, SCALE=400, BASE=10, INIT_RATING=1000, sample_weight=None
):
    from sklearn.linear_model import LogisticRegression
    ptbl_a_win = pd.pivot_table(
        df[df["winner"] == "model_a"],
        index="model_a",
        columns="model_b",
        aggfunc="size",
```



```

        fill_value=0,
    )
    # if no tie, create a zero matrix
    if sum(df["winner"].isin(["tie", "tie (bothbad)"])) == 0:
        ptbl_tie = pd.DataFrame(0, index=ptbl_a_win.index, columns=ptbl_a_win.
↪columns)
    else:
        ptbl_tie = pd.pivot_table(
            df[df["winner"].isin(["tie", "tie (bothbad)"])],
            index="model_a",
            columns="model_b",
            aggfunc="size",
            fill_value=0,
        )
        ptbl_tie = ptbl_tie + ptbl_tie.T

    ptbl_b_win = pd.pivot_table(
        df[df["winner"] == "model_b"],
        index="model_a",
        columns="model_b",
        aggfunc="size",
        fill_value=0,
    )
    ptbl_win = ptbl_a_win * 2 + ptbl_b_win.T * 2 + ptbl_tie

    models = pd.Series(np.arange(len(ptbl_win.index)), index=ptbl_win.index)

    p = len(models)
    X = np.zeros([p * (p - 1) * 2, p])
    Y = np.zeros(p * (p - 1) * 2)

    cur_row = 0
    sample_weights = []
    for m_a in ptbl_win.index:
        for m_b in ptbl_win.columns:
            if m_a == m_b:
                continue
            # if nan skip
            if math.isnan(ptbl_win.loc[m_a, m_b]) or math.isnan(ptbl_win.
↪loc[m_b, m_a]):
                continue
            X[cur_row, models[m_a]] = +math.log(BASE)
            X[cur_row, models[m_b]] = -math.log(BASE)
            Y[cur_row] = 1.0
            sample_weights.append(ptbl_win.loc[m_a, m_b])

            X[cur_row + 1, models[m_a]] = math.log(BASE)

```

```

        X[cur_row + 1, models[m_b]] = -math.log(BASE)
        Y[cur_row + 1] = 0.0
        sample_weights.append(ptbl_win.loc[m_b, m_a])
        cur_row += 2
    X = X[:cur_row]
    Y = Y[:cur_row]

    lr = LogisticRegression(fit_intercept=False, penalty=None, tol=1e-6)
    lr.fit(X, Y, sample_weight=sample_weights)
    elo_scores = SCALE * lr.coef_[0] + INIT_RATING
    if "mixtral-8x7b-instruct-v0.1" in models.index:
        elo_scores += 1114 - elo_scores[models["mixtral-8x7b-instruct-v0.1"]]
    return pd.Series(elo_scores, index=models.index).
↪sort_values(ascending=False)

```

```

[21]: elo_mle_ratings = compute_mle_elo(battles)
      pretty_print_model_ratings(elo_mle_ratings)

```

/home/madison/.mypipenv/lib/python3.14/site-packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was deprecated in version 1.8 and will be removed in 1.10. To avoid this warning, leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and C=np.inf instead of penalty=None.

```
warnings.warn(
```

```

[21]:
      Model  Elo rating
1      chatgpt-4o-latest  1315.16
2      gemini-1.5-pro-exp-0801  1298.43
3      gpt-4o-2024-05-13  1285.83
4      gpt-4o-mini-2024-07-18  1274.85
5      claude-3-5-sonnet-20240620  1270.98
..      ...
125     chatglm-6b  878.99
126     fastchat-t5-3b  868.29
127     stablelm-tuned-alpha-7b  839.69
128     dolly-v2-12b  822.08
129     llama-13b  798.92

```

[129 rows x 2 columns]

3.6.3 Compute Bootstrap Confidence Interavals for BT Scores

We can further use bootstrap to estimate the confidence intervals as well.

```

[22]: def get_bootstrap_result(battles, func_compute_elo, num_round):
      rows = []
      for i in tqdm(range(num_round), desc="bootstrap"):

```

```

        rows.append(func_compute_elo(battles.sample(frac=1.0, replace=True)))
df = pd.DataFrame(rows)
return df[df.median().sort_values(ascending=False).index]

```

[23]: BOOTSTRAP_ROUNDS = 100

```

np.random.seed(42)
bootstrap_elo_lu = get_bootstrap_result(battles, compute_mle_elo,
↳BOOTSTRAP_ROUNDS)

```

```

bootstrap:  0%|
| 0/100 [00:00<?, ?it/s]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:  1%|
| 1/100 [00:04<07:32,  4.57s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:  2%|
| 2/100 [00:08<07:18,  4.48s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:  3%|
| 3/100 [00:13<07:04,  4.38s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:  4%|
| 4/100 [00:17<06:54,  4.32s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

```

```

C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 5%|
| 5/100 [00:21<06:51, 4.33s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 6%|
| 6/100 [00:26<06:43, 4.29s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 7%|
| 7/100 [00:30<06:34, 4.25s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 8%|
| 8/100 [00:34<06:37, 4.32s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 9%|
| 9/100 [00:38<06:28, 4.27s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 10%|
| 10/100 [00:43<06:24, 4.27s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

```

```

C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 11%|
| 11/100 [00:47<06:16, 4.23s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 12%|
| 12/100 [00:51<06:14, 4.26s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 13%|
| 13/100 [00:55<06:06, 4.21s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 14%|
| 14/100 [00:59<06:00, 4.19s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 15%|
| 15/100 [01:04<05:58, 4.21s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 16%|
| 16/100 [01:08<05:51, 4.18s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

```

```

C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 17%|
| 17/100 [01:12<05:41, 4.12s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 18%|
| 18/100 [01:16<05:33, 4.07s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 19%|
| 19/100 [01:20<05:28, 4.05s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 20%|
| 20/100 [01:24<05:22, 4.04s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 21%|
| 21/100 [01:28<05:17, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 22%|
| 22/100 [01:32<05:13, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

```

```

C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 23%|
| 23/100 [01:36<05:09, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 24%|
| 24/100 [01:40<05:04, 4.01s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 25%|
| 25/100 [01:44<05:03, 4.04s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 26%|
| 26/100 [01:48<04:58, 4.04s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 27%|
| 27/100 [01:52<04:54, 4.03s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 28%|
| 28/100 [01:56<04:49, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

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C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 29%|
| 29/100 [02:00<04:45, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 30%|
| 30/100 [02:04<04:39, 4.00s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 31%|
| 31/100 [02:08<04:35, 4.00s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 32%|
| 32/100 [02:12<04:32, 4.01s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 33%|
| 33/100 [02:16<04:28, 4.01s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 34%|
| 34/100 [02:20<04:25, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

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C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 35%|
| 35/100 [02:24<04:22, 4.03s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 36%|
| 36/100 [02:28<04:17, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 37%|
| 37/100 [02:32<04:12, 4.01s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 38%|
| 38/100 [02:36<04:08, 4.00s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 39%|
| 39/100 [02:40<04:03, 3.99s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 40%|
| 40/100 [02:44<04:00, 4.01s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

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C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 41%|
| 41/100 [02:48<03:56, 4.01s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 42%|
| 42/100 [02:52<03:52, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 43%|
| 43/100 [02:56<03:49, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 44%|
| 44/100 [03:00<03:44, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 45%|
| 45/100 [03:04<03:40, 4.01s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 46%|
| 46/100 [03:08<03:35, 3.99s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

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C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 47%|
| 47/100 [03:12<03:33, 4.02s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 48%|
| 48/100 [03:16<03:29, 4.03s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 49%|
| 49/100 [03:20<03:25, 4.03s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 50%|
| 50/100 [03:24<03:21, 4.04s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 51%|
| 51/100 [03:28<03:20, 4.09s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 52%|
| 52/100 [03:32<03:14, 4.06s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

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C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 53%|
| 53/100 [03:37<03:14, 4.13s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 54%|
| 54/100 [03:41<03:14, 4.22s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 55%|
| 55/100 [03:45<03:09, 4.22s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 56%|
| 56/100 [03:50<03:10, 4.33s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 57%|
| 57/100 [03:54<03:07, 4.37s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 58%|
| 58/100 [03:59<03:02, 4.35s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

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C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 59%|
| 59/100 [04:03<02:54, 4.26s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 60%|
| 60/100 [04:07<02:47, 4.19s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 61%|
| 61/100 [04:11<02:42, 4.16s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 62%|
| 62/100 [04:15<02:41, 4.24s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 63%|
| 63/100 [04:20<02:38, 4.29s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 64%|
| 64/100 [04:24<02:36, 4.34s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and

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C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 65%|
| 65/100 [04:28<02:31, 4.33s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 66%|
| 66/100 [04:33<02:26, 4.32s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 67%|
| 67/100 [04:37<02:23, 4.34s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 68%|
| 68/100 [04:41<02:19, 4.35s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
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l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
69%|
| 69/100 [04:46<02:13, 4.32s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
70%|
| 70/100 [04:50<02:10, 4.35s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,

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leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
71%|
| 71/100 [04:55<02:07, 4.40s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
72%|
| 72/100 [04:59<02:01, 4.35s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
73%|
| 73/100 [05:03<01:58, 4.39s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
74%|
| 74/100 [05:08<01:54, 4.42s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
75%|
| 75/100 [05:12<01:48, 4.33s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.

```

```

    warnings.warn(
bootstrap:
76%|
| 76/100 [05:16<01:43, 4.31s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
77%|
| 77/100 [05:20<01:37, 4.25s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
78%|
| 78/100 [05:24<01:32, 4.21s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
79%|
| 79/100 [05:29<01:28, 4.21s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
80%|
| 80/100 [05:33<01:24, 4.23s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap:
81%|

```



```

| 81/100 [05:37<01:20, 4.24s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 82%|

| 82/100 [05:41<01:16,
4.24s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 83%|

| 83/100 [05:46<01:12,
4.24s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 84%|

| 84/100 [05:50<01:07,
4.20s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 85%|

| 85/100 [05:54<01:03,
4.22s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 86%|

| 86/100 [05:58<00:58,
4.20s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,

```

leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and C=np.inf instead of penalty=None.

```
warnings.warn(  
bootstrap: 87%|
```

```
| 87/100 [06:02<00:54,  
4.20s/it]/home/madison/.mypipenv/lib/python3.14/site-  
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was  
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,  
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use  
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and  
C=np.inf instead of penalty=None.
```

```
warnings.warn(  
bootstrap: 88%|
```

```
| 88/100 [06:07<00:50,  
4.22s/it]/home/madison/.mypipenv/lib/python3.14/site-  
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was  
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,  
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use  
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and  
C=np.inf instead of penalty=None.
```

```
warnings.warn(  
bootstrap: 89%|
```

```
| 89/100 [06:11<00:46,  
4.23s/it]/home/madison/.mypipenv/lib/python3.14/site-  
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was  
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,  
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use  
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and  
C=np.inf instead of penalty=None.
```

```
warnings.warn(  
bootstrap: 90%|
```

```
| 90/100 [06:15<00:42,  
4.22s/it]/home/madison/.mypipenv/lib/python3.14/site-  
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was  
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,  
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use  
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and  
C=np.inf instead of penalty=None.
```

```
warnings.warn(  
bootstrap: 91%|
```

```
| 91/100 [06:19<00:38,  
4.23s/it]/home/madison/.mypipenv/lib/python3.14/site-  
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was  
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,  
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use  
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and  
C=np.inf instead of penalty=None.
```

```

warnings.warn(
bootstrap: 92%|
| 92/100 [06:24<00:34,
4.26s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
warnings.warn(
bootstrap: 93%|
| 93/100 [06:28<00:29,
4.26s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
warnings.warn(
bootstrap: 94%|
| 94/100 [06:32<00:25,
4.27s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
warnings.warn(
bootstrap: 95%|
| 95/100 [06:36<00:21,
4.24s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
warnings.warn(
bootstrap: 96%|
| 96/100 [06:41<00:16,
4.24s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
warnings.warn(
bootstrap: 97%|
| 97/100 [06:45<00:12,

```

```

4.28s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 98%|
          | 98/100 [06:49<00:08,
4.29s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 99%|
          | 99/100 [06:54<00:04,
4.30s/it]/home/madison/.mypipenv/lib/python3.14/site-
packages/sklearn/linear_model/_logistic.py:1135: FutureWarning: 'penalty' was
deprecated in version 1.8 and will be removed in 1.10. To avoid this warning,
leave 'penalty' set to its default value and use 'l1_ratio' or 'C' instead. Use
l1_ratio=0 instead of penalty='l2', l1_ratio=1 instead of penalty='l1', and
C=np.inf instead of penalty=None.
    warnings.warn(
bootstrap: 100%|
          | 100/100 [06:58<00:00, 4.18s/it]

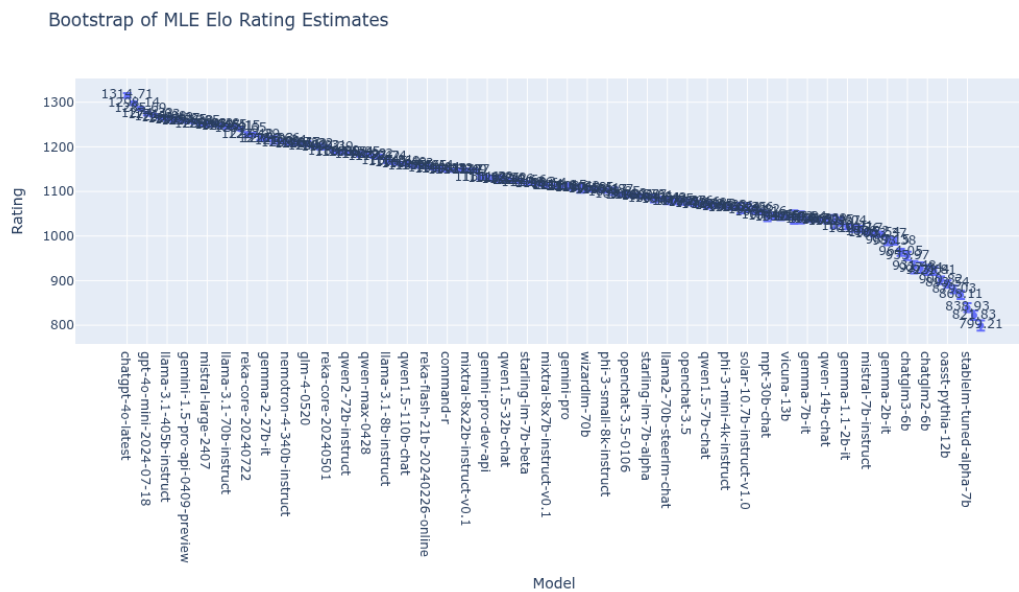
```

```

[24]: def visualize_bootstrap_scores(df, title):
    bars = pd.DataFrame(dict(
        lower = df.quantile(.025),
        rating = df.quantile(.5),
        upper = df.quantile(.975))).reset_index(names="model").
    ↪sort_values("rating", ascending=False)
    bars['error_y'] = bars['upper'] - bars["rating"]
    bars['error_y_minus'] = bars['rating'] - bars["lower"]
    bars['rating_rounded'] = np.round(bars['rating'], 2)
    fig = px.scatter(bars, x="model", y="rating", error_y="error_y",
                     error_y_minus="error_y_minus", text="rating_rounded",
                     title=title)
    fig.update_layout(xaxis_title="Model", yaxis_title="Rating",
                      height=600)
    return fig

fig = visualize_bootstrap_scores(bootstrap_elo_lu, "Bootstrap of MLE Elo Rating_
↪Estimates")
fig

```



We previously apply bootstrapping on the online Elo to obtain stabler ratings.

```
[25]: np.random.seed(42)
bootstrap_online_elo = get_bootstrap_result(battles, compute_online_elo,
BOOTSTRAP_ROUNDS)
```

```
bootstrap: 100%|
            | 100/100 [05:02<00:00, 3.02s/it]
```

We can see the bootstrapping medians obtained by both methods are similar.

```
[26]: preety_print_two_ratings(bootstrap_elo_lu.quantile(.5),
                                bootstrap_online_elo.quantile(.5),
                                column_names=["Bootstrap Median of BT", "Bootstrap
                                ↪Median of Online Elo"])
```

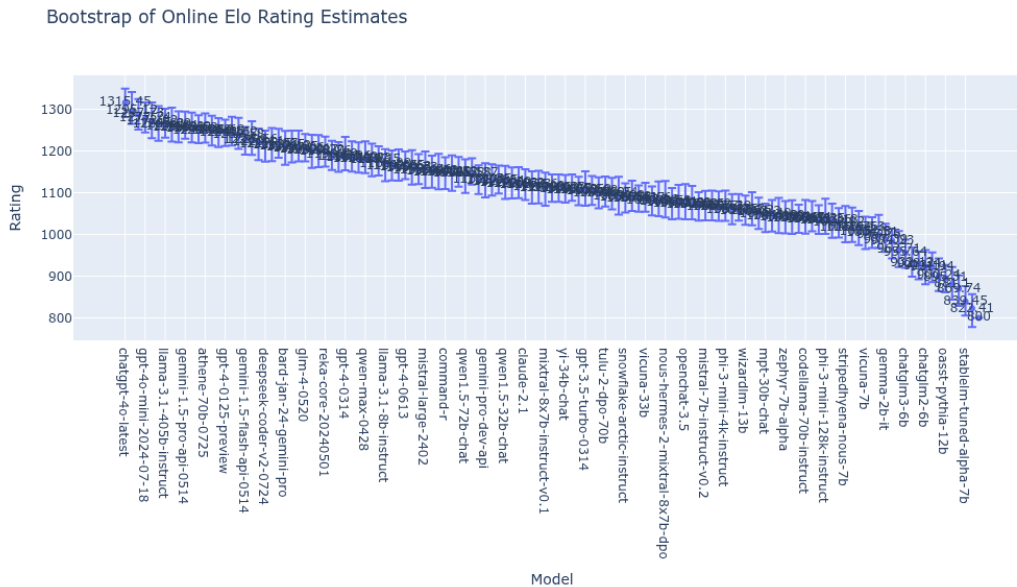
[26] :	Model	Bootstrap Median of BT \
1	chatgpt-4o-latest	1315
2	gemini-1.5-pro-exp-0801	1298
3	gpt-4o-2024-05-13	1286
4	gpt-4o-mini-2024-07-18	1274
5	claude-3-5-sonnet-20240620	1271
..	...	...
125	chatglm-6b	879
126	fastchat-t5-3b	868
127	stablelm-tuned-alpha-7b	839
128	dolly-v2-12b	822
129	llama-13b	799

	Bootstrap Median of Online Elo
1	1315
2	1295
3	1288
4	1278
5	1272
..	...
125	881
126	870
127	839
128	822
129	800

[129 rows x 3 columns]

However, online Elo's confidence intervals are significantly larger than the BT.

```
[27]: fig = visualize_bootstrap_scores(bootstrap_online_elo, "Bootstrap of Online Elo_
↳Rating Estimates")
fig
```



3.6.4 Predict Win Rates

Utilizing Elo ratings allows us to predict win probabilities. By comparing the predicted win rates with the actual win rates, we can gain insight into the accuracy and quality of the Elo rating system.

```
[28]: def predict_win_rate(elo_ratings, SCALE=400, BASE=10, INIT_RATING=1000):
    names = sorted(list(elo_ratings.keys()))
    wins = defaultdict(lambda: defaultdict(lambda: 0))
    for a in names:
        for b in names:
            ea = 1 / (1 + BASE ** ((elo_ratings[b] - elo_ratings[a]) / SCALE))
            wins[a][b] = ea
            wins[b][a] = 1 - ea

    data = {
        a: [wins[a][b] if a != b else np.NAN for b in names]
        for a in names
    }

    df = pd.DataFrame(data, index=names)
    df.index.name = "model_a"
    df.columns.name = "model_b"
    return df.T
```

```
[29]: win_rate = predict_win_rate(dict(bootstrap_elo_lu.quantile(0.5)))
ordered_models = win_rate.mean(axis=1).sort_values(ascending=False).index
ordered_models = ordered_models[:30]
fig = px.imshow(win_rate.loc[ordered_models, ordered_models],
                color_continuous_scale='RdBu', text_auto=".2f",
                title="Predicted Win Rate Using Elo Ratings for Model A in an A_
↳vs. B Battle")
fig.update_layout(xaxis_title="Model B",
                  yaxis_title="Model A",
                  xaxis_side="top", height=900, width=900,
                  title_y=0.07, title_x=0.5)
fig.update_traces(hovertemplate=
                  "Model A: %{y}<br>Model B: %{x}<br>Win Rate: %{z}<extra></
↳extra>")
fig
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[29], line 1
----> 1 win_rate = predict_win_rate(dict(bootstrap_elo_lu.quantile(0.5)))
      2 ordered_models = win_rate.mean(axis=1).sort_values(ascending=False).index
      3 ordered_models = ordered_models[:30]

Cell In[28], line 11, in predict_win_rate(elo_ratings, SCALE, BASE, INIT_RATING
      7         wins[a][b] = ea
      8         wins[b][a] = 1 - ea
     10 data = {
----> 11     a: [wins[a][b] if a != b else np.NAN for b in names]
```

```

12     for a in names
13 }
15 df = pd.DataFrame(data, index=names)
16 df.index.name = "model_a"

File ~/mypipenv/lib/python3.14/site-packages/numpy/__init__.py:792, in
↳ __getattr__(attr)
    789     import numpy.char as char
    790     return char.chararray
--> 792 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'NaN'

```

3.6.5 Compute Bootstrap Confidence Intervals Assuming Uniform Sampling

We also study how the ratings will change if we only sample an equal number of battles for each model pair.

```

[ ]: def sample_battle_even(battles, n_per_battle):
    groups = battles.groupby(["model_a", "model_b"], as_index=False)
    resampled = (groups
                  .apply(lambda grp: grp.sample(n_per_battle, replace=True))
                  .reset_index(drop=True))
    return resampled

```

```

[ ]: num_samples = 50
    battles_even = sample_battle_even(battles, num_samples)
    pd.pivot_table(battles_even, index="model_a", columns="model_b",
↳ aggfunc="size", fill_value=0)

```

```

[ ]: # Sampling Battles Evenly
def get_bootstrap_even_sample(battles, n_per_battle, func_compute_elo,
↳ num_round=BOOTSTRAP_ROUNDS):
    rows = []
    for n in tqdm(range(num_round), desc="sampling battles evenly"):
        resampled = sample_battle_even(battles, n_per_battle)
        rows.append(func_compute_elo(resampled))
    df = pd.DataFrame(rows)
    return df[df.median().sort_values(ascending=False).index]

```

```

[ ]: print("number of samples per battle pair:", num_samples)
    bootstrap_even_lu = get_bootstrap_even_sample(battles, num_samples,
↳ compute_mle_elo, num_round=100)

```

```

[ ]: fig = visualize_bootstrap_scores(bootstrap_even_lu, f"Bootstrap of MLE Elo
↳ Estimates - Even sample")
    fig

```


4 Language-specific Leaderboards

We present two language-specific leaderboards, by isolating the chat data into two subsets based on the language: (1) English-only and (2) Non-English.

4.1 English-only

```
[ ]: english_only_battles = battles[battles["language"] == "English"]
     elo_ratings = compute_mle_elo(english_only_battles)
     pd.DataFrame(elo_ratings)
```

4.2 Non-English

```
[ ]: non_english_battles = battles[battles["language"] != "English"]
     elo_ratings = compute_mle_elo(non_english_battles)
     pd.DataFrame(elo_ratings)
```

5 Links

Some good resources to learn more about Elo rating systems: - Elo rating system https://en.wikipedia.org/wiki/Elo_rating_system - Bradley-Terry model https://en.wikipedia.org/wiki/Bradley%E2%80%93Terry_model - An introduction video <https://www.youtube.com/watch?v=AsYfbmp0To0> - A FiveThirtyEight article <https://fivethirtyeight.com/methodology/how-our-nfl-predictions-work/>